1 Fig. 1.1 shows a horizontal turntable.

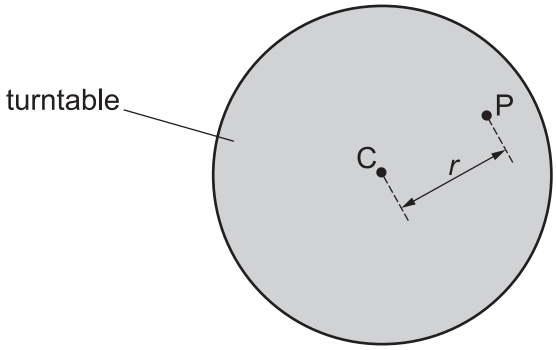

Fig. 1.1

Point C is at the centre of the turntable. Point P is a distance r from the centre.

Fig. 1.2 shows a side view of a d.c. motor attached to the turntable with a belt.

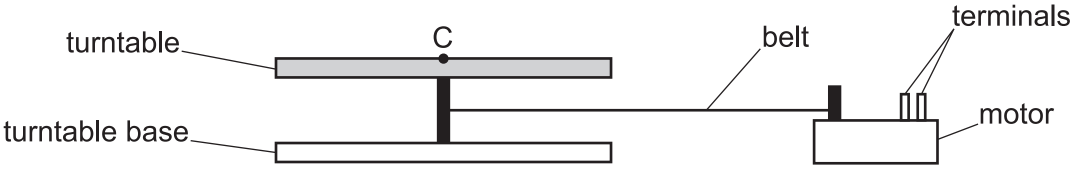

Fig. 1.2

The motor is used to rotate the turntable at frequency f_0. The motor is switched off and the turntable continues to rotate at frequency f_0.

A sphere of adhesive putty of mass m is dropped onto the turntable at point P. The frequency of the turntable is now f.

It is suggested that f is related to m by the relationship

$$\frac{Kf_0}{f} = \beta K + mr^2$$

where β and K are constants.

Plan a laboratory experiment to test the relationship between f and m.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for β and K.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

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